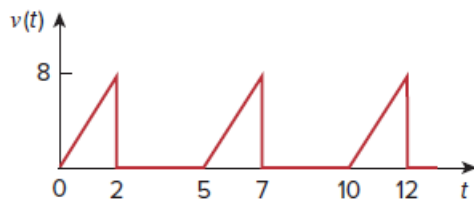


## Problem

Find the rms value of the voltage waveform of Fig. 11.59 as well as the average power absorbed by a  $2\text{-}\Omega$  resistor when the voltage is applied across the resistor.

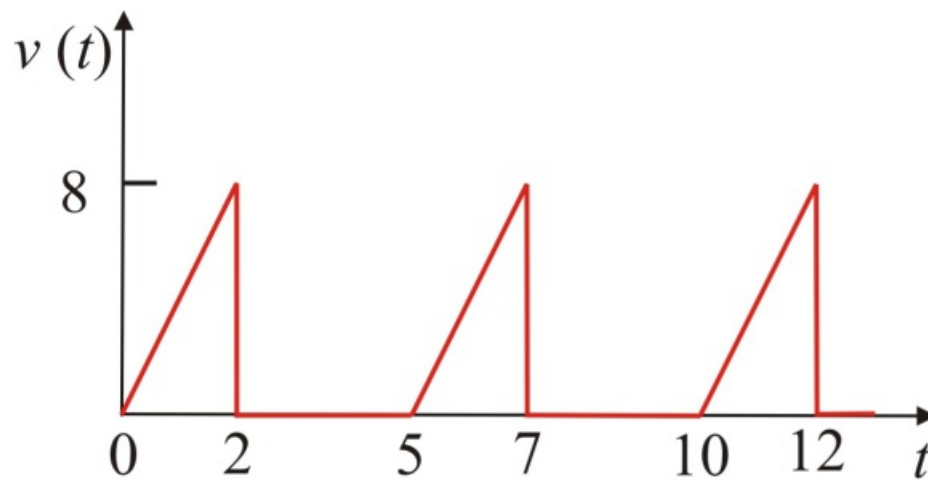
**Figure 11.59:**



## Step-by-step solution

## Step 1 of 3

Draw the voltage waveform  $v(t)$  as shown in Figure 1.



**Figure 1**

Express  $v(t)$  in mathematical form in one period of time.

$$v(t) = \begin{cases} 4t, & 0 \leq t < 2 \\ 0, & 2 \leq t < 5 \end{cases}$$

## Step 2 of 3

Determine the rms value of the voltage.

$$\begin{aligned} V_{\text{rms}}^2 &= \frac{1}{T} \int_0^T v^2(t) dt \\ &= \frac{1}{(5-0)} \left[ \int_0^2 (4t)^2 dt + \int_2^5 (0)^2 dt \right] \\ &= \frac{1}{5} \left[ 16 \frac{t^3}{3} \Big|_0^2 + 0 \right] \\ &= \frac{1}{5} \left[ \frac{16}{3} (2^3 - 0^3) \right] \end{aligned}$$

Simplify further.

$$\begin{aligned} V_{\text{rms}}^2 &= \frac{128}{15} \\ V_{\text{rms}} &= 2.92 \text{ V} \end{aligned}$$

Thus, the rms value of the voltage waveform is  $\boxed{2.92 \text{ V}}$ .

### Step 3 of 3

Write the expression of the average power absorbed by  $2 \Omega$  resistor.

$$P = \frac{V_{\text{rms}}^2}{R}$$

Substitute  $2 \Omega$  for  $R$  and  $2.92 \text{ V}$  for  $V_{\text{rms}}$ .

$$\begin{aligned} P &= \frac{2.92^2}{2} \\ &= 4.26 \text{ W} \end{aligned}$$

Thus, the average power absorbed by  $2 \Omega$  resistor is  $\boxed{4.26 \text{ W}}$ .